A3-PS015-TC120

Explainable Home and Last-Mile Connectivity Diagnosis

Prof. Sudip Barik, Pushon Bhattacharyya, Tuhin Mondal
sudip.barik@tint.edu.in , pushonbhattacharyya590@gmail.com , tuhinmondal036@gmail.com
AIORI Hackathon 3
October 12, 2026

Problem Description

The Problem

Conventional speed tests measure peak throughput under idle conditions, overlooking loaded latency, jitter, and packet loss across fiber, 5G, and satellite networks. Consequently, users experience lag and interruptions despite high bandwidth, leading to misdiagnosed faults, consumer frustration, and ineffective troubleshooting.

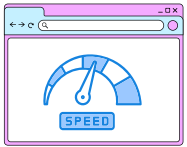
Key Issue:-

- **Failure Domain Blindness (Wi-Fi vs. ISP):** Standard tests measure end-to-end performance in one shot, which intern cannot distinguish local Wi-Fi issues from ISP or last-mile failures.
- **Metric Mismatch:** Tests focus almost entirely on peak throughput, completely ignoring the metrics that dictate app stability: DNS resolution time, packet loss, and loaded jitter.
- **Medium Disregard:** Fiber, cellular (5G), fixed wireless, and satellite are treated under identical benchmarks, masking medium-specific traits like radio contention, weather fading, and inherent latency floors.
- **Unexplainable Fault Attribution:** Because idle tests report passing speeds, neither the User nor the ISP can diagnose intermittent drops—leading to false complaints, misdirected support tickets, and unresolved network degradation.

Background

Why high bandwidth doesn't always mean reliable connectivity.

SECTION 1



Speed ≠ Quality

"High bandwidth can still mean lag, buffering, and interrupted streamings."

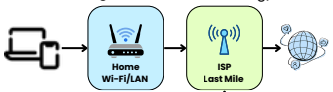
- Latency
- Jitter
- Packet Loss
- DNS Delay

SECTION 2

Where is the Fault?

Home Network (Wi-Fi, router, local congestion)

External Network (ISP access link, routing)



"End-to-end tests often cannot isolate where degradation begins."

SECTION 3

The Proposed Approach

1. Measure
Test idle and loaded conditions.

2. Diagnose
Compare metrics and isolate likely fault domains.

3. Recommend
Guide the next appropriate action.

RFC 2330-inspired performance measurement

Literature & RFC Review

Standards, frameworks and studies most relevant to **TicNet**

STANDARDS



RFC 2330: IP Performance Metrics Framework

Conceptual basis for consistent, reproducible measurement.



RFC 2679/7679 & 2681: One-Way & Round-Trip Delay

Defines how to measure one-way andround-trip delay.



RFC 2680/7680: Packet Loss

Defines how to measure one-way packet loss.



RFC 3393: Delay Variation (Jitter)

Standardizes measurement of packet delay variation.

STUDIES



Sundaresan et al.: Home Network or Access Link?

Tells home Wi-Fi problems apart from ISP problems. Our core idea.



Gettys & Nichols: Bufferbloat

Why delay rises when the network is busy.

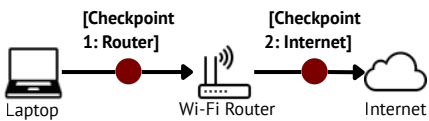
Proposed Solution

TicNet: Explainable Connectivity Diagnosis

A guided desktop agent that finds where the internet is failing, and why.

1. **Measure** — Delay, loss, jitter, DNS and speed, idle and busy.
2. **Localise** — Compare the router with the internet.
3. **Explain** — Simple rules with a confidence score and the numbers behind it.
4. **Escalate** — If unsure, say 'inconclusive' and create report for a technician.

The two-point test



Problem at Checkpoint 1 → Wi-Fi or home network

Clean at 1, problem at 2 → ISP or last mile

How TicNet decides

Evidence	Diagnosis
Loss or jitter at the router	Local Wi-Fi / LAN fault
Router clean, loss beyond it	Last-mile / ISP loss
Path clean, slow DNS	DNS problem
Delay jumps when busy	Congestion(bufferbloat)

Optimization Proposed

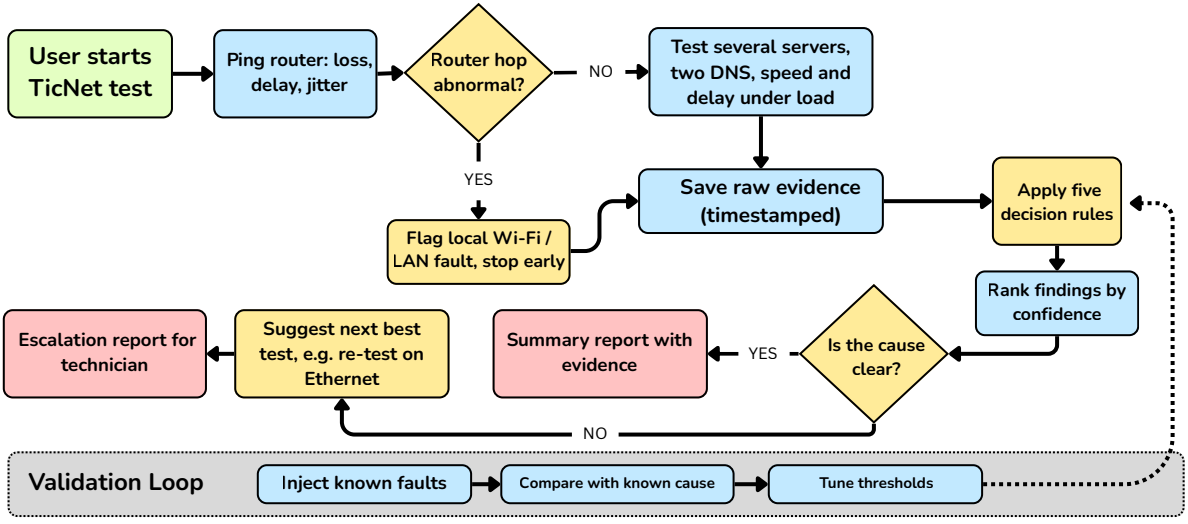
What makes TicNet Smarter and Lighter

Six parameters where TicNet improves on existing speed tests

Upgrade	Before: TicNet	After: TicNet
Staged testing	Runs every test every time	Stops early once the fault is found
Random sampling	One short burst of probes	Random probes over time, median and 95th percentile
Cross-checks	One server, one DNS	Several servers and a second DNS
Smart baselines	One fixed limit for everyone	Adapts to your own normal and connection type
Unclear cases	Escalate straight away	Suggests the next best test first, e.g. re-test on Ethernet
Impact score	Raw numbers only	Calls, gaming and streaming: OK or poor

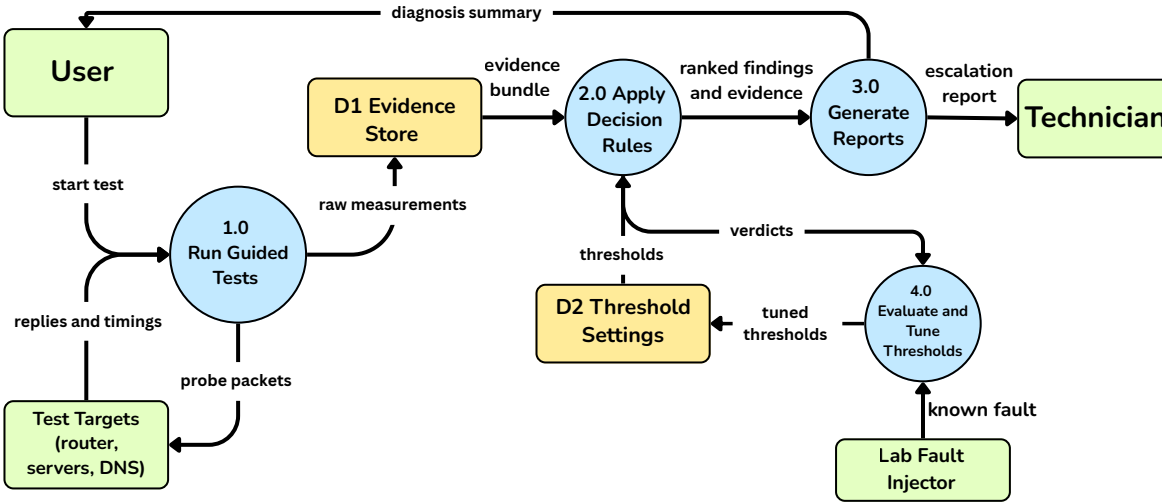
Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.



Data Flow Diagram

How Data Moves Through TicNet



Deployment Timeline

Our Projected Execution Plan

1

Design
Weeks 1–2

Study the problem, plan the system.

2

Build
Weeks 3–4

Code the agent, rules and reports.

3

Test
Weeks 5–6

Break the network on purpose, check TicNet's answers.

4

Pilot and Demo
Weeks 5–6

Try it with a few friends, write the report.

Methodology

1. Measure
Follow RFC 2330. Save every result as timestamped evidence.

→

2. Inject faults
Wi-Fi jitter, packet loss, slow DNS, congestion.

→

3. Compare
Does TicNet name the cause we injected?

→

4. Tune and repeat
Adjust thresholds, run the tests again.

Bibliography

Four key papers that have been studied for TicNet

1 

Measuring Internet Speed: Current Challenges and Future Recommendations

- Feamster & Livingood, Communications of the ACM, 2020
- Used for: why one speed number is not enough. Click Here

2 

Home Network or Access Link? Locating Last-Mile Downstream Throughput Bottlenecks

- Sundaresan, Feamster, Teixeira, PAM, 2016
- Used for: telling a home Wi-Fi problem from an ISP problem. Click Here

3 

Bufferbloat: Dark Buffers in the Internet

- Gettys & Nichols, Communications of the ACM, 2012
- Used for: testing delay under load, not only when idle. Click Here

4 

A Survey on Internet Performance Measurement Platforms and Related Standardization Efforts

- Bajpai & Schönwälder, IEEE Communications Surveys & Tutorials, 2015
- Used for: large-scale probe platforms, which supports our volunteer pilot. Click Here

Thank You

TicNet